

MOVING CONTINENTS

THE CONTINENTS THAT MAKE UP most of the Earth's land surface are always on the move, shifted around by forces deep inside the Earth. This is known as continental drift. Movement, or drift, takes place because of intense heat generated within the Earth. The heat is carried upward where it disturbs the cool, rocky surface, or crust, forcing sections of it, called plates, to move. Each year the continents, parts of the plates, drift nearly half an inch (about a centimeter), some getting closer together, others moving farther apart, some grinding past each other. As this happens, many of the Earth's natural features are created or changed.

IN THE BEGINNING

Scientists believe that some 300 million years ago all the land on Earth was joined together in one "supercontinent" called Pangaea. It was surrounded by a giant ocean, Panthalassa. About 200 million years ago, as the plates moved, Pangaea began to split into two great landmasses, Laurasia in the north, and Gondwanaland in the south. These were separated by the Tethys Sea. As the plates continued to move, the two landmasses split and moved farther apart, eventually forming the continents on the map below.



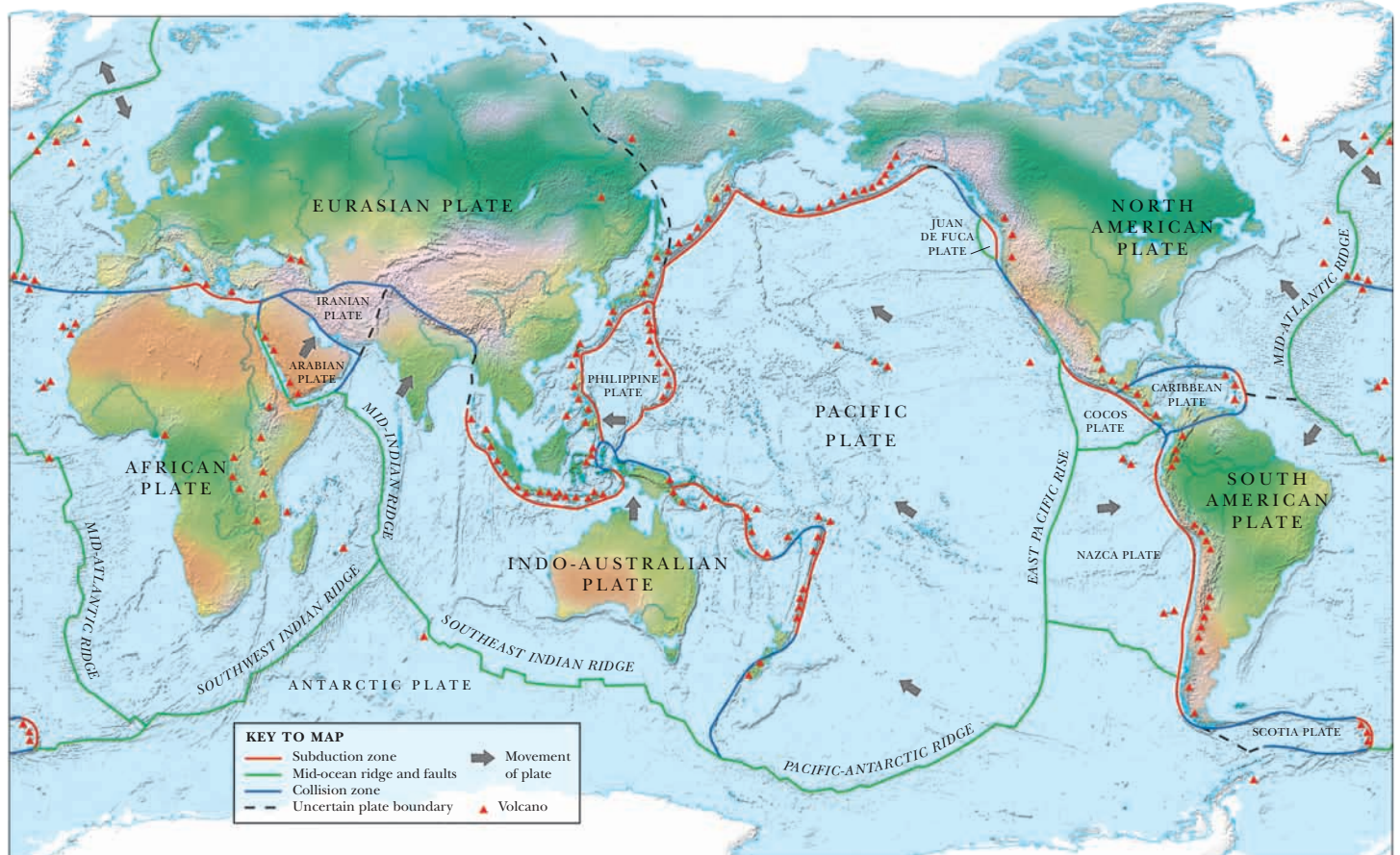
200 million years ago



180 million years ago

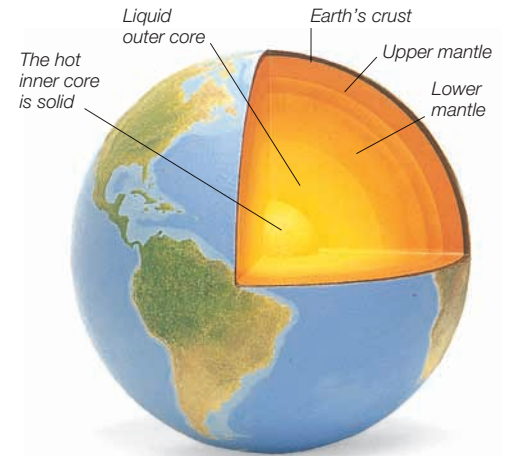


65 million years ago



KEY TO MAP

- Subduction zone
- Mid-ocean ridge and faults
- Collision zone
- - - Uncertain plate boundary
- ➔ Movement of plate
- ▲ Volcano



INSIDE THE EARTH

The Earth is not a solid ball, but is made up of many different layers. The crust that forms the continents and the ocean floors is a thin layer of rock that covers the Earth like a shell. The mantle beneath is 1,864 miles (3,000 km) thick and made of hot rock, some of which is molten (liquid). At the center is the core, the hot metallic center of the Earth. This is liquid on the outside and solid on the inside.